

Flavio Giuliani

Ph.D. in Physics

Current address & contacts

Dipartimento di Fisica, Sapienza University of Rome

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- My profile on scientific databases: [Google Scholar](#), [ResearchGate](#), [ORCID](#)
- My github profile: github.com/flagiu
- My website: flagiu.github.io

Education

- 2022 - 2026: Ph.D. in Physics, XXXVIII ciclo, ottimo con lode (magna cum laude), Sapienza University of Rome.
 - Thesis: *Study of the thermodynamics and kinetic anomalies of Antimony-based phase-change materials by machine-learned molecular dynamics*, supervised by Prof. Riccardo Mazzarello and Prof. Lilia Boeri
 - The thesis project involved the study of thermodynamic and kinetic anomalies and crystallization features in supercooled liquid Sb, a candidate phase-change material, and the development of a neural network interatomic potential trained on density functional theory for Ge-alloyed Sb
- 2020 - 2022: Master's Degree in Physics, 110/110 e lode (magna cum laude), Sapienza University of Rome.
 - Thesis: *Vibrational dynamics of simulated glasses with different fictive temperatures*, supervised by Prof. Tullio Scopigno and Prof. Francesco Sciortino
 - The aim of the thesis project was to assess the correlation between supercooled liquid viscosity and glass vibrational properties for a prototype fragile glass former: the rigid model of ortho-terphenyl
- 2017 - 2020: Bachelor's Degree in Physics, 110/110 e lode (magna cum laude), Sapienza University of Rome
 - Thesis: *Twitching motility in bacteria*, supervised by Prof. Roberto Di Leonardo
 - The thesis project consisted in reproducing a tug-of-war physical model for simulating the twitching dynamics of a bacterium on a surface, mediated by the adhesion and retraction of type-IV pili

Publications

I have published 2 articles in international peer-reviewed scientific journals. I am first author of 1 of those publications, on the high-Impact Factor (IF) journal "Proceedings of the National Academy of Sciences U.S.A." (PNAS U.S.A., IF: 9.1).

1. F. Giuliani, F. G. Mattioli, Y. Chen, D. Baratella, D. Dragoni, M. Bernasconi, J. Russo, L. Boeri & R. Mazzarello, "Liquid anomalies and fragility of supercooled antimony", PNAS U.S.A. 123 (12) e2531605123, DOI: 10.1073/pnas.2531605123 (2026)
2. S. Ritarossi, R. Piombo, F. Giuliani, D. Dragoni, M. Bernasconi, R. Mazzarello, "Phase-Change Heterostructures Based on Antimony", *Phys. Status Solidi RRL* **19** (7), 2500012 (2025)

Teaching

- 2024: Teaching Assistant for "Mathematics and Statistics", Bachelor's Degree in Environmental Sciences, Sapienza University of Rome. Interactive exercises and homework corrections (40 h)
- 2023: 1 h lecture "Machine learning interatomic potentials" for the course "Computing Methods for Physics" of Master's Degree in Physics by Prof. L. Boeri, Sapienza University of Rome

- 2021, 2022: Student-collaboration scholarship for Scientific Programming in C (“Laboratorio di Calcolo”), Bachelor’s Degree in Physics, Sapienza University of Rome. Laboratory assistant and notes-writer (75 h/year)

Certified advanced and soft-skill courses

- 2024: Course “[HPC Molecular Modelling](#)”, three-days lessons and hands-on at CINECA, Rome
- 2022 - 2023: Within my Ph.D. education, Sapienza University of Rome
 - “Statistical Field Theory” (40 h, 6 credits) by Prof. A. Cavagna, 30/30 e lode (magna cum laude)
 - “Advanced Topics in Condensed Matter” (40 h, 6 credits) by Prof. J. Lorenzana and Dr. and Dr. P. Barone, 30/30 con lode (magna cum laude)
 - Soft-skills course “Preparing artwork for scientific papers” by Prof. G. Giardina
 - Soft-skills online courses “I principali programmi europei di finanziamento R&I”, “Etica e Scienza”, “Fare start-up in Sapienza, istruzioni per l’uso” and “Publications and research data: Open Access”

Grants

High-Performance Computing (HPC) resources

- 2025: 20k GPUh grant as Principal Investigator (P.I.) of the [ISCRA-C project](#) “LIGAN: Liquid-liquid transition in Germanium-doped Antimony with a Neural-network interaction potential” for the Leonardo supercomputer
- 2024: 20k GPUh grant as P.I. of the [ISCRA-C project](#) “AntimoNN: Neural-Network interaction potential for Antimony-based phase-change materials” for the Leonardo supercomputer
- 2023: 23333 CPUh grant as P.I. of the [ISCRA-C project](#) “DAGAN: First-principles DATaset of Germanium-Antimony for a Neural-network interaction potential” for the G100 supercomputer

Other grants

- 2025: €11k grant for the University call for Starting Third Mission initiatives, Sapienza University of Rome
 - Title: *Eutopie - Dialogues on the relationship between science and society for a research on human beings*
 - Role: Proponent and Head of Organization. I oversaw the entire organizational cycle: format design, selection of interdisciplinary topics, research and involvement of 15+ high-profile experts, logistics management, communication, and feedback analysis through questionnaires.
 - Objective and implementation: The project created an open space for multidisciplinary debate in the university to address major contemporary social challenges, promoting collective involvement through four public dialogues on: 1) Man, nature, and the climate crisis; 2) Science, women, and rights; 3) Human language and artificial intelligence; 4) Research and imagination in art and science.

Awards

- 2022: [Excellence program scholarship](#) of Master’s Degree in Physics, Sapienza University of Rome
 - Awarded to top 15 students based on exam marks
- 2020: [Excellence program scholarship](#) of Bachelor’s Degree in Physics, Sapienza University of Rome
 - Awarded to top 34 students based on exam marks

Conferences & seminars

Contributions

- 2024: Poster presentation at “[CMT@BRIXEN ’24: The Meeting of the Condensed Matter Theory Italian Community](#)”, Brixen, Italy.
 - Poster: [Ab initio study of supercooled antimony alloys through a machine learning interatomic potential](#)

- 2024: 10' talk at the workshop "Multiscale Modeling and Engineering Applications", ICSC - National Research Center in HPC, Big Data and Quantum Computing - Spoke 6, Sapienza University of Rome
 - Talk: "Neural-network interatomic potential for antimony-based phase-change materials", within the Flagship Project FP1 - Advanced Materials and New Devices
- 2023: 20' PhD Journal club seminar of the Physics Department, Sapienza University of Rome
 - Talk: "Accessing large-scale molecular simulations through machine learning"

Participant

- 2024: Participant at "European Phase-Change and Ovonic Symposium" (E\PCOS2024), Leibniz Institute of Surface Engineering (IOM), Leipzig, Germany
- 2023: Participant and Technical staff at E\PCOS2023, Sapienza University of Rome
- 2022: Participant at Workshop "From Water to Colloidal Water - Celebrating the contributions of Francesco Sciortino to Soft Matter", Sapienza University of Rome
- 2021: Participant at "XXVII INC Summer School: Ultrastable Glasses - New perspectives for an old problem", organized by Instituto Nicolás Cabrera, Universidad Autónoma de Madrid, in "La Cristalera", Miraflores de la Sierra, Madrid, Spain
- 2020: Participant at "Summer School: Physics of Life", organized by the Center for the Physics of Biological Function, Princeton University, online

Scientific communication for a broad audience

- 2025: Talk for the book presentation "L'atomica e le responsabilità della scienza, organized by L'Asino d'oro Edizioni at Libreria Spazio Sette, Rome.

Volunteering

- 2019: Co-founder of "STudy to transfORM" (ST.ORM), a nonprofit organization whose educational projects currently span across Italy, Kenya and Tanzania.
- 2019: For about one year, I taught math and sciences to refugees within the activities of the non-profit organization "Baobab Experience" in Rome.

Personal skills

- Language: Italian (native), English (B2), Spanish (A2)
- Driving license: Italian B